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A network intrusion detection method based on contrastive learning and Bayesian Gaussian Mixture Model

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Abstract

Network Intrusion Detection Systems (NIDS) are essential for safeguarding networks against malicious activities. However, existing machine learning-based NIDS often require complex feature engineering, which demands significant domain expertise and experimentation, leading to suboptimal model performance in complex network environments. In contrast, deep learning approaches, while powerful, struggle with imbalanced data, resulting in a bias towards normal traffic and reduced effectiveness in detecting rare attacks. To address these issues, we propose a method that combines contrastive learning and Bayesian Gaussian Mixture Model (BGMM). Specifically, we propose a novel contrastive learning loss that enables the model to automatically learn the similarity within normal traffic and the distinction between normal and malicious traffic, thereby generating robust and distinguishable feature representations. This approach not only eliminates the need for manual feature engineering but also helps alleviate the issue of weak feature representations for rare attacks. BGMM further enhances detection performance by adapting to both normal and malicious patterns through the use of multiple components. The effectiveness of the proposed method is validated through extensive experiments on two widely used modern network intrusion datasets. On the UNSW-NB15 dataset, the proposed method achieves 91.27% accuracy and 92.30% F1-score, which is 1.85% and 2.35% better than the state-of-the-art (SOTA) method. On the Distrinet-CIC-IDS2017 dataset, the proposed method achieves 99.66% accuracy and 99.12% F1-score, which is 0.05% and 0.12% better than the SOTA method.

Keywords Cybersecurity, Network intrusion detection, Contrastive learning, Bayesian Gaussian Mixure Model

Introduction

The development of network technologies such as 5 G and cloud computing has greatly enhanced people's lives by providing faster and more reliable connectivity. However, this rapid advancement has also transformed the digital landscape, making it more susceptible to cyber threats. As a result, there has been a significant increase in cyberattacks, with attackers employing increasingly sophisticated and covert methods that complicate detection efforts. This challenge is underscored by a report jointly released by Google's Threat Analysis Group (TAG) and Mandiant, which revealed that the number of exploited zero-day vulnerabilities reached 97 in 2023, an increase of over 50% from 62 in 2022.[1] Such a surge in vulnerabilities highlights the evolving threat landscape and emphasizes the urgent need for effective defense mechanisms.

In this context, Network Intrusion Detection Systems (NIDS) play a crucial role, as they can promptly detect malicious activities and safeguard the security and stability of systems (Ahmad et al. 2021). By effectively identifying these sophisticated threats, NIDS help organizations

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[1] https://storage.googleapis.com/gweb-uniblog-publish-prod/documents/Year_in_Review_of_ZeroDays.pdf

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mitigate risks associated with advanced cyberattacks and protect their digital infrastructure.

NIDS can be categorized into signature-based and anomaly-based systems. While signature-based detection methods achieve high accuracy for known threats, they often fail to identify novel attacks, such as zero-day vulnerabilities (Jin et al. 2021; Sommestad et al. 2022), exposing organizations to significant risks. In contrast, anomaly-based detection methods analyze network traffic to identify deviations from established the normal pattern, enabling the detection of previously unseen threats (Yang et al. 2022). Thus, this paper focuses on anomaly-based NIDS because of their superior capability to identify emerging threats, making them crucial for addressing the dynamic landscape of cybersecurity.

To implement anomaly-based detection effectively, various machine learning models have been employed, including Decision Tree Classifier (DTC) (Guezzaz et al. 2021; Mahbooba et al. 2021), Random Forest (RF) (Balyan et al. 2022), Support Vector Machine (SVM) (Turukmane and Devendiran 2024), and XGBoost (Jiang et al. 2020). However, these models encounter a significant challenge: their heavy dependence on feature engineering. The need for manual feature selection and extraction complicates the model training process, necessitating domain expertise and extensive experimentation to identify effective features (Zhou et al. 2020; Disha and Waheed 2022; Li et al. 2024). This reliance can lead to suboptimal model performance, especially in complex network environments where traffic data exists in high-dimensional forms.(Kasongo and Sun 2020).

Recognizing these limitations, researchers have turned to deep learning methods, which have gained traction in network intrusion detection due to their powerful feature extraction capabilities and ability to handle complex data. However, deep learning approaches also face unique challenges, particularly in the context of anomaly detection. Firstly, these models typically require large amounts of labeled data for effective training. In network intrusion detection, malicious activities are relatively rare, resulting in severely imbalanced datasets. This imbalance can cause models to favor the majority class (usually normal traffic), thereby diminishing their ability to detect minority class instances (anomalous traffic or attacks) (He et al. 2023; Hu et al. 2021; Su et al. 2020).

Moreover, while techniques such as SMOTE (Zhang et al. 2020), GAN (Andresini et al. 2021; Huang and Lei 2020), VAE (Liu et al. 2022), Diffusion Model (Tang et al. 2023), and ADASYN Abdelkhalek and Mashaly 2023 have been proposed to tackle class imbalance, they introduce new challenges, including increased training overhead and reliance on the quality of generated samples. These factors further limit the effectiveness of deep learning methods in anomaly detection.

In light of these challenges, this paper presents a novel two-stage network intrusion detection approach that combines contrastive learning with Bayesian Gaussian Mixture Models (BGMM). The first stage leverages contrastive learning to automatically learn representations that enhance the ability to distinguish between normal and abnormal traffic. By automating the representation learning process, our method reduces the reliance on manual feature engineering, while the distinguishable features between normal and malicious traffic enhance the detection rate of rare attacks. In the second stage, we introduce a decision algorithm based on BGMM for anomaly detection. BGMM adapts to different traffic patterns by allowing for a variable number of components, which enhances the detection performance of complex network traffic.

To summarize, the key contributions of this paper can be outlined as follows:

(1) We present an innovative two-stage network intrusion detection method that integrates a representation module utilizing contrastive learning with a detection module based on Bayesian Gaussian Mixture Model (BGMM), effectively addressing complex network attacks.
(2) A novel supervised contrastive learning loss function is introduced to automate the learning of representations that distinguish between normal and abnormal traffic. This approach reduces reliance on manual feature engineering and effectively addresses the imbalanced data challenges in deep learning, enhancing the detection of rare attacks.
(3) The anomaly detection algorithm leverages BGMM, allowing for a flexible number of components to adapt to each traffic pattern, which enhances the algorithm's performance by improving its flexibility and robustness in detecting anomalies.
(4) To validate the effectiveness of our proposed method, extensive experiments are conducted on two benchmark datasets, including UNSW-NB15 and Distrinet-CIC-IDS2017, demonstrating performance improvements in detection.

The rest of the paper is organized as follows: Sect. "Related work" provides an overview of related work. Section "Methodology" details the proposed method, outlining its key components and mechanisms. Section "Results and discussion" presents the experimental validation, where the performance of the proposed method is evaluated using benchmark datasets. Finally, Sect. "Conclusion" concludes the paper by summarizing

the contributions of the proposed method and discussing potential avenues for the future research.

Related work

In this part, an overview of recent research on intrusion detection systems that employ machine learning, deep learning and contrastive learning is presented.

Recent studies in intrusion detection utilizing machine learning have highlighted the importance of designing effective feature selection techniques. Yin et al. (2023) proposed the IGRF-RFE approach, which integrated information gain and random forest filtering with recursive feature elimination, successfully reducing the number of features from 42 to 23 and improving multilayer perceptron (MLP) accuracy from 82.25% to 84.24% on the UNSW-NB15 dataset. Halim et al. (2021) introduced a GA-based feature selection method (GbFS) that employed optimized parameters and a novel fitness function. When tested on the CIRA-CIC-DOHBrw-2020 and UNSW-NB15 datasets, GbFS demonstrated improved performance, achieving a maximum accuracy of 99.80%. Wu and Li (2021) developed an ensemble of neural networks and random forest to identify feature subsets, showcasing enhanced detection performance on the KDD99 and Honeynet datasets. Albulayhi et al. (2022) proposed an innovative feature selection and extraction method for IoT intrusion detection systems, utilizing entropy-based techniques, which resulted in the identification of up to 28 relevant features on the IoTID20 dataset and 25 features on the NSL-KDD dataset, with a classification accuracy of 99.98%. Collectively, these studies underscore the critical role of effective feature selection in addressing the challenges posed by high-dimensional data in machine learning-based intrusion detection systems.

Network intrusion detection leveraging deep learning has garnered significant interest due to its capability to automatically extract meaningful features and effectively recognize complex patterns. For instance, Halbouni et al. (2022) proposed a method that combines Convolutional Neural Networks (CNN) with Long Short-Term Memory networks (LSTM) to capture the spatial-temporal features of network traffic. Their results on the CIC-IDS 2017, UNSW-NB15, and WSN-DS datasets demonstrated a high detection rate with relatively low false alarm rates (FAR).

Similarly, Altunay and Albayrak (2023) explored the performance of CNN, LSTM, and the hybrid CNN-LSTM approach. They found that the CNN-LSTM model achieved the best detection performance, with an accuracy of 93.21% for binary classification and 92.9% for multi-class classification on the UNSW-NB15 dataset. Furthermore, this model attained a detection accuracy of 99.84% for binary classification and 99.80% for multi-class classification in the X-IIoTID dataset.In another study, Dai et al. (2024) developed a method integrating CNN, Bidirectional LSTM (BiLSTM), and an attention mechanism to enhance the key components of spatial-temporal features extracted by the CNN-BiLSTM framework, thereby improving detection performance. They also employed equalization loss to mitigate the impact of class imbalance.

Several studies have addressed the challenges posed by class imbalance in network traffic, which can degrade the performance of deep learning models. For instance, Fu et al. (2022) utilized Adaptive Synthetic Sampling (ADASYN) to construct a balanced dataset. Their model, similar to Dai et al. (2024), combined CNN-BiLSTM with an attention mechanism, achieving accuracies of 90.73% and an F1 score of 89.65% on the NSL-KDD dataset. Additionally, Lee and Park (2021) employed Generative Adversarial Networks (GAN) to generate samples of minority classes and utilized Random Forest for detection, demonstrating superior results compared to training directly on the imbalanced dataset. Moreover, Liu et al. (2022) implemented Variational Autoencoders (VAE), Conditional VAE, and Random Undersampling combined with Conditional VAE to generate minority class samples, using CNN for detection. Their experiments on the CSE-CIC-IDS2018 dataset verified the effectiveness of the three data processing schemes.

In recent years, contrastive learning has led to significant advancements across various fields, particularly in enhancing model performance through the acquisition of more meaningful feature representations. By maximizing the similarity between positive pairs and minimizing it between negative pairs, contrastive learning enables models to capture subtle patterns in high-dimensional data. This approach has proven especially impactful in applications such as computer vision (Li et al. 2022), natural language processing (Gao et al. 2021), speech recognition (Wang et al. 2022b), and text prediction (An et al. 2022).

Based on the characteristics of contrastive learning, some studies have applied it to the field of intrusion detection, exploring how to leverage this method to enhance detection performance. Some of these studies are outlined below:

Wang et al. (2022a) introduced an intrusion detection system for IoT devices that combines contrastive learning with federated learning. They performed feature selection and used contrastive learning to create representation vectors from processed data. Anomalies were detected by comparing the cosine similarity score of the test sample against a fixed threshold based on the average of normal representation vectors.

Yue et al. (2022) developed a dual-branch network intrusion detection system integrating cross-entropy and supervised contrastive learning loss functions. They created positive sample pairs using random masking and processed feature vectors through two branches: one for prediction labels and cross-entropy loss, and another for representation vectors and contrastive loss. The total loss was optimized through backpropagation, with the category obtained from the prediction branch for anomaly detection.

Li et al. (2024) proposed an end-to-end system that preprocesses raw network traffic and uses CNN and GRU networks to extract spatiotemporal features. These features are converted into representation vectors, and contrastive loss is calculated to optimize the network. Anomaly detection was performed using a fixed threshold similar to Wang et al. (2022a).

Yuan et al. (2024) designed an intrusion detection system for 5 G networks, using both raw and statistical traffic features to create vectors. They formed positive and negative sample pairs based on transport layer protocols and trained autoencoders with a combination of reconstruction and contrastive loss. Anomaly detection utilized the Median Absolute Deviation (MAD) method with a voting mechanism.

Zhang et al. (2024) presented an online intrusion detection system with an anomaly detection module and a pseudo-label generation module. The anomaly detection module enhanced autoencoder representation using a new contrastive loss function. Two Gaussian distributions based on cosine similarity scores were used for anomaly detection, with larger means indicating normal patterns. Results were determined through voting, and pseudo-labels for test samples were generated for ongoing learning.

Methodology

The framework of the proposed network intrusion detection method for binary classification of network traffic samples, which combines contrastive learning and Bayesian Gaussian Mixture Model, is shown in Fig. 1.

In the train stage, a multilayer perceptron network with a contrastive learning loss function is trained to obtain the representation vectors of network traffic samples. After having trained the network, the normal template is acquired by averaging the representation vectors of all normal samples, finally the cosine similarity scores between the representation vectors and the normal template are calculated to train Bayesian Gaussian Mixture Model for anomaly detection. In the test stage, the network traffic sample under test passes through the representation module to obtain its representation vector, and then the cosine similarity score between the representation vector and the normal template is computed. Finally, the label of the sample is determined by comparing the posterior probabilities of various components.

In the following, the two main modules of the proposed method are explained in more detail.

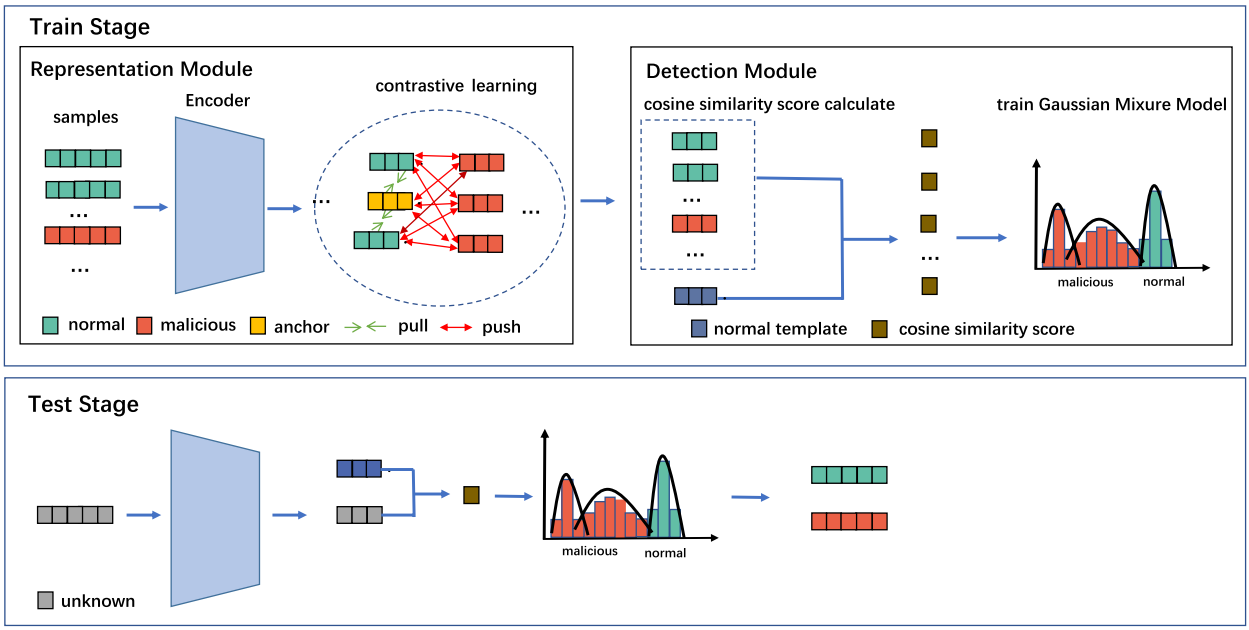


Fig. 1 Framework of the proposed method

Representation module based on contrastive learning

In the representation module, a multi-layer perceptron encoder with a contrastive learning loss function is used for the generation of representation vectors. Contrastive learning methods typically perturb samples slightly to construct positive pairs, while other samples serve as negative pairs. These constructed positive and negative pairs are used to generate discriminative representation vectors. In the intrusion detection, the goal is to distinguish between normal network activities and malicious network activities. Traffic samples inherently possess positive and negative characteristics, namely normal traffic samples and malicious traffic samples. Applying minor perturbations may obscure the characteristics of the samples, leading to a decrease in the performance of the intrusion detection systems. Therefore, identical to the method used by Wang et al. (2022a) and Zhang et al. (2024), we construct positive and negative pairs by using normal samples as positive instances and malicious samples as negative instances.

The dataset $D = \{X, Y\}$ is assumed, where $X = \{x_1, x_2, \ldots, x_i, \ldots, x_{N+M}\}$, with $x_i \in \mathbb{R}^n$, represents $N + M$ traffic samples. In this set, N samples are normal and M samples are malicious. The corresponding labels are denoted by $Y = \{y_1, y_2, \ldots, y_i, \ldots, y_{N+M}\}$, where $y_i \in \{0, 1\}$. Here, 0 denotes normal samples and 1 denotes malicious ones.

The sample x_i is transformed by the encoder to obtain a low-dimensional representation vector $v_i = f_\theta(x_i)$ belonging to $\mathbb{R}^d$, where θ are the trainable parameters of the encoder. In the d-dimensional space, we aim to bring the representation vectors of positive samples closer while distancing them from those of negative samples, thereby enhancing the distinguishability between normal and malicious representation vectors. When considering repulsive relationships, previous methods either focus solely on the relationship between a single anchor sample and negative samples, neglecting the relationship between other positive samples and negative samples, or concentrate exclusively on the relationship between all positive samples and negative samples, overlooking the relationship between positive samples and the anchor sample. To address these limitations, we propose a novel contrastive learning loss function that simultaneously accounts for the relationships between the anchor sample and positive samples, as well as between positive samples and negative samples. The differences among the contrastive learning loss functions of different methods are shown in Fig. 2.

In this paper, the similarity between two vectors is measured using cosine similarity score, as detailed in Eq. 1.

$$h(i,j) = \frac{v_i^T \cdot v_j}{\|v_i\| \|v_j\|} \tag{1}$$

When constructing negative sample pairs, we not only iterate through all positive samples and all negative samples, but also take into account the relationship between each positive sample and the anchor sample. This adjustment ensures that the representation vectors of different classes maintain a certain level of distinguishability, while also making the representation vectors of positive samples cluster more closely together.

The proposed contrastive learning loss function is formulated as:

$$w(i,j) = \frac{1}{exp(h(i,j)/\beta)} \tag{2}$$

$$t(i,j) = exp(h(i,j)/\tau) \tag{3}$$

$$\mathscr{L}_{i,j} = -log \frac{t(i,j)}{t(i,j) + \sum_{l=1}^{N} w(i,l) \sum_{k=1}^{M} t(l,k)} \tag{4}$$

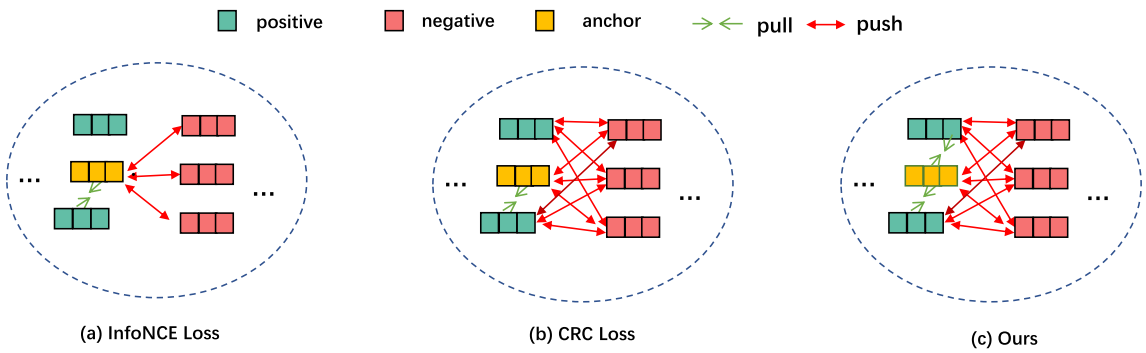


Fig. 2 Different contrastive learning loss functions used in intrusion detection field. **a** InfoNce Loss used in Feco (Wang et al. 2022a), **b** CRC Loss used in AOC-IDS (Zhang et al. 2024), **c** Ours

$$\mathscr{L} = \frac{1}{N(N-1)} \sum_{i=1}^{N} \sum_{j=1, j\neq i}^{N} \mathscr{L}_{i,j} \tag{5}$$

where $\tau \in (0,1]$ is the temperature parameter and β is a hyperparameter used to prevent the encoder from focusing too much on aggregating positive samples with the anchor sample, neglecting the repulsion between positive samples and negative samples.

Detection module based on Bayesian Gaussian Mixure Model

In this section, a brief overview of the Bayesian Gaussian Mixture Model (BGMM)(Bishop and Nasrabadi 2006) and variational inference is provided, followed by an illustration of how this model can be utilized for anomaly detection.

Bayesian Gaussain mixure model and variational inference

Gaussian Mixture Model (GMM) is a kind of probabilistic model that combines multiple Gaussian densities, each contributing to the overall probability density function with its own weight. A d-dimensional Gaussian density, also called the component,is described as follows:

$$\mathscr{E}\ (x|\mu, \Sigma) = \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right) \tag{6}$$

$$N(x|\mu, \Sigma) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \cdot \mathscr{E}\ (x|\mu, \Sigma) \tag{7}$$

where x is a d-dimensional vector, μ belongs to $\mathbb{R}^d$, and Σ is a $d \times d$ matrix representing the mean and covariance matrix of the Gaussian density respectively.

A d-dimensional Gaussian Mixture Model consisting of K components is described as follows:

$$f_K(x) = \sum_{k=1}^{K} w_k N(x|\mu_k, \Sigma_k) \tag{8}$$

where w_k is the weight of the k-th component, with the requirement that $\sum_{k=1}^{K} w_k = 1$, and $w_k \geq 0$.

The parameters, $\Theta = \{\theta_1, \theta_2, \ldots, \theta_K | \theta_k = \{w_k, \mu_k, \Sigma_k\}\}$ in Eq. 8, need to be estimated to use GMM for representing data distributions. The Expectation-Maximization (EM) algorithm(Dempster et al. 1977) is typically employed for this purpose. The algorithm iteratively maximizes the likelihood of GMM through the expectation step and the maximization step until convergence. However, the EM algorithm requires pre-specifying the number of mixture components and may converge to local optima (Zimek et al. 2012).

To address these issues, we employ the Bayesian Gaussian Mixture Model (BGMM), which utilizes variational inference to approximate the posterior distributions of the model. Compared to the GMM, the BGMM introduces priors on the parameters of the GMM. Specifically, the conjugate prior over $\{\mu_k, \Sigma_k\}$ for each Gaussian component is given by the Gaussian-inverse-Wishart distribution:

$$(\mu_k, \Sigma_k) \sim \mathcal{N}(\mu_k|m_0, \Sigma_k/\beta_0) \times \mathcal{W}^{-1}(\Sigma_k|\Psi_0, \nu_0) \tag{9}$$

where m_0 is the mean of the Gaussian prior, β_0 is the scale factor of the Gaussian prior, Ψ_0 is the scale matrix of the Wishart prior, and ν_0 is the degrees of freedom of the Wishart prior. For the weights $\mathbf{w} = \{w_1, \ldots, w_k\}$, the prior follows a Dirichlet distribution:

$$\mathbf{w} \sim \mathrm{Dir}(\alpha_0) \tag{10}$$

where α_0 is the concentration parameter of the Dirichlet prior.

By incorporating prior distributions, BGMM can dynamically adjust the model complexity to better fit the actual structure of the data. When an initial number of components is set for the BGMM, it automatically determines the effective number of components by adjusting the weights of some components close to zero. This approach not only avoids the need for manual specification of the number of components but also enhances the model's robustness against uncertainty and the risk of overfitting.

Variational inference (Blei et al. 2017) approximates the posterior distribution using a variational distribution. In simple terms, it transforms the problem of minimizing the Kullback-Leibler (KL) divergence between the variational distribution and the posterior distribution into maximizing a lower bound, known as the Evidence Lower Bound (ELBO).

In the BGMM, the ELBO can be expressed as:

$$\mathcal{L}(q) = \mathbb{E}_{q(\Theta)}[\log p(X, \Theta)] - \mathbb{E}_{q(\Theta)}[\log q(\Theta)] \tag{11}$$

where joint distribution

$$p(X, \Theta) = p(\mathbf{w}) \cdot p(X|\Theta) \cdot \prod_{k=1}^{K} p(\mu_k|m_0, \Sigma_k/\beta_0) \cdot p(\Sigma_k|\Psi_0, \nu_0) \tag{12}$$

and variational distribution

$$q(\Theta) = q(\mathbf{w}) \cdot \prod_{k=1}^{K} q(\mu_k|m_k, \Sigma_k/\beta_k) \cdot q(\Sigma_k|\Psi_k, \nu_k) \tag{13}$$

By maximizing the ELBO, we can obtain the optimal parameters for the model.To gain a more in-depth understanding of the BGMM, readers are encouraged to consult the textbook (Bishop and Nasrabadi 2006).

Bayesian Gaussian mixure model for anomaly detection

In the anomaly detection phase, previous studies mostly use a fixed threshold, which only detects simple anomalies and is sensitive to noise points. To address the issue, Zhang et al. (2024) proposed using two Gaussian distributions to represent the normal pattern and the malicious pattern respectively, and anomalies are detected by comparing the posterior probabilities of Gaussian distributions, which improves robustness of the detection method. However, the normal pattern and the malicious pattern may not be represented by simple unimodal distributions. Using only two Gaussian distributions cannot adequately capture the complexity of these patterns.

To address this issue, we employ the BGMM to characterize the patterns. BGMM, by utilizing the Dirichlet process, allows the number of components to adaptively adjust based on the complexity of the data, eliminating the need for prior specification. BGMM can utilize multiple components to characterize both the normal and the malicious patterns, thereby enhancing the robustness of the detection algorithm. Specifically, the cosine similarity scores between the representation vectors and the normal template are used for the training of BGMM. After training the BGMM, it is necessary to determine which components correspond to the normal pattern and which correspond to the malicious pattern. In this work, we assume that if a component contains more than 80% normal samples, it is classified as part of the normal pattern; otherwise, it is classified as part of the malicious pattern. During anomaly detection, the classification of samples as either normal or malicious is based on comparing the posterior probabilities of the different components. The anomaly detection algorithm based on BGMM is outlined in Algorithm 1.

Algorithm 1 Decision process based on BGMM

Input:
Representation vectors of the samples during training, v_{train};
Representation vector of the sample under test, v^i_{test};
Normal template, $\overline{v}$;
Trained Bayesian Gaussian Mixture Model with K components, Θ;
Threshold for normal components, $\alpha = 0.8$;
Output:
Predict label of the test sample, $\overline{y^i_{test}}$;

Calculate the cosine similarity scores between training representation vectors and the normal template, s_{train};
For each component c_k in BGMM, determine its normality:
for each component c_k in BGMM **do**
  Acquire all samples assigned to c_k based on their posterior probabilities:
  $N_k = \{x^j_{train} | \mathbb{I}(i = k) \arg\max_i P(i|s^j_{train}, \theta_i)\}$
  $i = 1, \ldots, K \quad j = 1, \ldots, N$;
  Calculate the proportion of normal samples in c_k:
  $\text{normal_ratio}_k = \frac{N_k^{normal}}{N_k}$
  if $\text{normal_ratio}_k \geq \alpha$ **then**
    Add component c_k to the normal pattern set: $\text{NC} \leftarrow \text{NC} \cup \{c_k\}$
  else
    Add component c_k to the malicious pattern set: $\text{MC} \leftarrow \text{MC} \cup \{c_k\}$
  end if
end for
For representation vector of the test sample v^i_{test}, calculate its cosine similarity score with the normal template, s^i_{test};
Determine the component with the highest posterior probability for the test sample: $c^i_{test} = \arg\max_k \{P(z = k|s^i_{test}, \theta_k)\} \quad k = 1, 2, \ldots, K$
Assign the label to the test sample based on its component:
if c^i_{test} *in* NC **then**
  $\overline{y^i_{test}} = 0$ // Classified as normal
else
  $\overline{y^i_{test}} = 1$ // Classified as malicious
end if
return $\overline{y^i_{test}}$

Results and discussion

In this section, we conduct extensive experiments using the UNSW-NB15 dataset (Moustafa and Slay 2015) and the Distrinet-CIC-IDS2017 dataset (Liu et al. 2022b), both of which are widely recognized benchmark datasets that reflect modern network attack characteristics. The proposed method is compared with various existing approaches based on accuracy, precision, recall, and F1-score, demonstrating its superiority.

Datasets description

Experiments are conducted on two benchmark datasets, the UNSW-NB15 dataset and the Distrinet-CIC-IDS2017 dataset. A detailed description of each is given below.

Details of UNSW-NB15 dataset

The UNSW-NB15 dataset was created and open-sourced by researchers at the the Cyber Range Lab of UNSW Canberra. It is designed to generate a blend of real-world contemporary normal activities and synthesized contemporary attack behavior. It primarily addresses the issue that previous datasets are too outdated to reflect modern network traffic scenarios, vast varieties of low footprint intrusions, and detailed structured information about network traffic. It should be noted that this dataset has the class imbalance problem, with the number of malicious samples in the training set being nearly twice that of the normal samples.The records in the UNSW-NB15 dataset can be categorized into 10 classes: Normal, Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shell code, and Worms. The number and distribution of each class, as described in Table 1.

Table 1 Data distribution of each class in the UNSW-NB15 dataset

Type		Number of records	
		Train set	Test set
Normal		56000	37000
Malicious	Analysis	2000	677
	Backdoor	1746	583
	DoS	12264	4089
	Exploits	33393	11132
	Fuzzers	18184	6062
	Generic	40000	18871
	Reconnaissance	10491	3496
	Shell code	1133	378
	Worms	130	44
Total		175341	82332

Details of distrinet-CIC-IDS2017 dataset

To address the limitations of insufficient traffic diversity and volume, inadequate representation of certain attack types, and anonymized packet data that fails to reflect current trends, researchers from the Canadian Institute for Cybersecurity (CIC) at the University of New Brunswick (UNB), Canada, created and released the CIC-IDS2017 dataset (Sharafaldin et al. 2018). However, other researchers later uncovered several previously overlooked issues during its development, including flaws in attack orchestration, feature generation, documentation, and labeling. In response to these issues, they thoroughly re-engineered the dataset, revised the labeling logic, and corrected the identified errors, ultimately leading to the release of the improved dataset, Distrinet-CIC-IDS2017. In this study, we use the updated dataset, splitting it into training and testing sets in an 80:20 ratio. It should be noted that this dataset also suffers from the class imbalance problem, with the number of normal samples in the training set being nearly four times that of the malicious samples. The records in the Distrinet-CIC-IDS2017 dataset are categorized into 16 classes: Normal, Botnet, DDoS, DoS-GoldenEye, DoS-Hulk, DoS-Slowhttptest, DoS-Slowloris, FTP-Patator, Heartbleed, Infiltration, Infiltration-Portscan, Portscan, SSH-Patator,Brute Force, SQL Injection, and XSS. The number and distribution of each class, as described in Table 2.

Table 2 Data distribution of each class in the distrinet-CIC-IDS2017 dataset

Type		Number of records	
		Train set	Test set
Normal		923528	231225
Malicious	Botnet	580	156
	DDoS	76173	18971
	DoS-goldenEye	6170	1397
	DoS-hulk	126876	31573
	DoS-slowhttptest	1394	347
	DoS-slowloris	3211	787
	FTP-patator	3188	784
	Heartbleed	7	4
	Infiltration	32	4
	Infiltration-portscan	4374	1111
	Portscan	1350	333
	SSH-patator	2344	617
	Brute force	60	13
	SQL injection	10	3
	XSS	15	3
Total		1149312	287328

Evaluation metrics

Various evaluation metrics, including accuracy, precision, recall, F1-score, and computational cost, are used to compare the proposed method with existing ones to prove its superiority. Malicious samples are considered positive, and normal samples are considered negative. True positive (TP) and true negative (TN) represent samples that are correctly classified, while false positive (FP) and false negative (FN) represent samples that are misclassified. The calculation methods for the evaluation metrics are as follows:

Accuracy

Accuracy represents the proportion of correctly classified samples in the entire dataset, calculated as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{14}$$

Precision

Precision represents the proportion of true positive samples among all samples classified as positive, calculated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{15}$$

Recall

Recall represents the proportion of true positive samples among all samples labeled as positive, calculated as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{16}$$

F1-score

F1-score is the harmonic mean of precision and recall, which simultaneously considers the accuracy and completeness of the classifier. F1-score can be calculated using the following formula:

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{17}$$

Computational cost

The computational cost refers to the time consumed during both the training and inference phases of the model. which can be expressed as:

$$C = T_{\text{train}} + T_{\text{inference}} \tag{18}$$

Data preprocess and experiment setup

The UNSW-NB15 dataset needs to be prepared to align with the model's input requirements, whereas the Distrinet-CIC-IDS2017 dataset can be used directly without additional modification. To prepare the dataset, one-hot encoding is applied to categorical fields, and min-max normalization is used for continuous attributes. As a result, the processed samples have the following dimensionalities: 196 dimensions for the UNSW-NB15 dataset, and 82 dimensions for the Distrinet-CIC-IDS2017 dataset. The experiments are conducted on Kaggle, a platform that offers data scientists an accessible and user-friendly environment for data analysis and model building. The hyperparameters used in the proposed method are presented in Table 3.

Evaluation of performance

Our method is compared with state-of-the-art NIDS models, including the deep learning-based CANET proposed by Ren et al. (2023), the spatiotemporal feature network CNN-LSTM-GRU called CTS proposed by Kilichev et al. (2024), the contrastive learning-based CIDS proposed by Yue et al. (2022), and AOC-IDS proposed by Zhang et al. (2024). Additionally, several widely used machine learning baselines, such as Support Vector Machine (SVM), Random Forest (RF), Decision Tree Classifier (DTC), AdaBoost, and XGBoost, are also included.

Evaluation of UNSW-NB15 dataset

The original partitioning method of the UNSW-NB15 dataset is employed to train and test the proposed model and baselines. The results are summarized in Table 4. From the table, it is evident that the proposed method excels in all metrics, and outshines baselines with the highest accuracy of 91.27% and the highest F1-score of 92.30%. Furthermore, a detailed confusion

Table 3 Hyperparameters of the proposed method

Hyperparameter	Dataset	
	UNSW-NB15	Distrinet-CIC-IDS2017
Epochs	200	10
Batch size	256	512
Encoder	(196,128–128,64)	(82,64–64,32)
Temperature	0.15	0.075
Beta	0.075	0.15
Inital Components	10	20
Activation	ReLU	ReLU
Optimizer	Adam	Adam
Learning Rate	0.001	0.001

Table 4 Performance(%) comparison between ours and baselines on the UNSW-NB15 dataset

Method	Evaluation metrics			
	Acc	Pre	Rec	F1
Ours	**91.27**	89.74	95.01	**92.30**
AOC-IDS	89.42	**94.30**	85.98	89.95
CIDS	82.18	75.90	99.10	85.96
CANET	87.91	83.18	97.82	89.91
CTS	84.42	78.40	98.99	87.50
SVM	81.61	75.11	**99.60**	85.64
DTC	84.51	80.03	95.76	87.19
RF	83.69	77.47	99.26	87.02
AdaBoost	82.73	76.63	98.77	86.30
XGBoost	83.58	77.57	98.72	86.88

The optimal metric performance is highlighted in bold

Table 5 Confusion matrix of the proposed method on the UNSW-NB15 dataset

Type		Predict Label	
		Normal	Malicious
True label	Normal	32075	4925
	Malicious	2264	43068

matrix for our method is provided in Table 5 to illustrate these results. From Table 4, it can be observed that while some methods, such as SVM, exhibit excellent recall (e.g., 99.60%), they perform poorly in terms of precision. This suggests that although these methods have a high detection rate for malicious samples, they also misclassify a substantial number of normal samples as malicious.

Furthermore, we use recall to evaluate the model's detection capability across different sample categories, especially for minority classes. The results are presented in Table 6. From the table, it can be seen that for minority attack types, such as Analysis,Worms Backdoor, the proposed method achieves a high detection rate.

In addition to performance metrics, it is essential to consider the computational overhead associated with each method. While our approach shows notable detection capabilities, it is also crucial to evaluate the efficiency and scalability of the model. We use Eq. 18 to measure the computational overhead, and the results are presented in the table below.

Table 7 presents the computational costs, in terms of training (T_{train}) and testing time (T_{test}), for various methods on the UNSW-NB15 dataset. Notably, our method demonstrates a balanced computational efficiency, with moderate training time and a low testing time. In contrast, methods like SVM exhibit high testing costs (510.41s), which could limit their practicality in real-time applications.

Evaluation of distrinet-CIC-IDS2017 dataset

The evaluation results of our model on the Distrinet-CIC-IDS2017 dataset are shown in Table 8. As seen in Table 8, our approach demonstrates a well-balanced performance across all evaluation metrics, underscoring its effectiveness in distinguishing both normal and malicious samples. The detailed confusion matrix is provided in Table 9.

Notably, most models achieve relatively high performance on this dataset. This may be due to the distinctive features present in the Distrinet-CIC-IDS2017 dataset, which, despite the class imbalance issue, make it easier for models to differentiate between classes.

Table 6 Recall rate(%) of each type between ours and baselines on the UNSW-NB15 dataset

Type	Method									
	Ours	AOC-IDS	CIDS	CANET	CTS	SVM	DTC	RF	XGBoost	AdaBoost
Normal	86.69	**93.63**	61.45	75.76	66.58	59.57	70.73	64.63	65.03	63.09
Analysis	99.41	93.21	99.85	**100.00**	**100.00**	99.85	94.39	**100.00**	96.75	98.87
Backdoor	99.49	96.05	**100.00**	99.83	**100.00**	**100.00**	98.11	**100.00**	99.83	99.66
DoS	99.14	92.54	99.83	**99.95**	99.93	**99.95**	98.24	99.88	99.88	99.54
Exploits	97.87	91.43	99.85	99.49	99.63	**99.95**	97.28	99.79	99.81	99.60
Fuzzers	68.54	29.81	93.88	85.05	93.32	**97.15**	78.75	94.92	91.41	92.23
Generic	99.90	99.05	**100.00**	99.96	99.99	**100.00**	99.24	**100.00**	99.98	99.96
Reconnai-ssance	98.57	85.58	99.71	99.57	**99.97**	**99.97**	98.71	99.97	99.89	99.89
Shell code	97.35	78.31	99.47	99.47	99.74	**100.00**	94.71	**100.00**	99.47	99.47
Worms	**100.00**	84.09	97.73	**100.00**	**100.00**	**100.00**	97.73	**100.00**	**100.00**	**100.00**

The optimal metric performance and the minority class are highlighted in bold

Table 7 Computational cost(\s) comparison between ours and baselines on the UNSW-NB15 dataset

Method	Computational cost	
	T_{train}	T_{test}
Ours	1028.10	0.32
AOC-IDS	1197.76	3.92
CIDS	1234.11	1.52
CANET	1853.45	7.93
CTS	673.13	10.46
SVM	1622.66	510.41
DTC	**4.84**	**0.04**
RF	37.04	1.28
AdaBoost	40.16	2.17
XGBoost	14.43	0.18

The optimal metric performance is highlighted in bold

Table 8 Performance(%) comparison between ours and baselines on the Distrinet-CIC-IDS2017 dataset

Method	Evaluation metrics			
	Acc	Pre	Rec	F1
Ours	**99.66**	98.77	99.47	**99.12**
AOC-IDS	96.24	83.85	**99.97**	91.20
CIDS	99.18	97.44	98.40	97.92
CANET	99.07	**99.76**	95.44	97.55
CTS	98.91	97.65	96.74	97.20
SVM	98.50	98.93	93.33	96.05
DTC	98.92	98.73	95.69	97.19
RF	99.29	99.55	96.80	98.16
AdaBoost	99.61	99.02	98.99	99.00
XGBoost	99.57	99.27	98.52	98.89

The optimal metric performance is highlighted in bold

Table 9 Confusion matrix of the proposed method on the Distrinet-CIC-IDS2017 dataset

Type		Predict label	
		Normal	Malicious
True label	Normal	230530	695
	Malicious	296	55807

AOC-IDS achieved the highest recall rate, however, it also misclassified a large number of normal samples as malicious. This may be because their detection method relies on only two Gaussian distributions to represent normal and malicious patterns, respectively, which overlooks the diversity within each pattern and ultimately leads to poorer overall performance.

In the same way, we assess the model's detection performance across different sample categories in the Distrinet-CICIDS2017 dataset using recall, focusing especially on minority classes. The findings are shown in Table 10. The table shows that for most attack categories, our method achieves a high detection rate. However, for minority attacks, such as Heartbleed and Infiltration, the detection rate is low due to the extremely limited number of samples in the training set.

The computational overhead of the proposed method and baseline models is presented in Table 11. As can be observed, our method demonstrates acceptable training and testing time overhead compared to other approaches. Notably, the SVM-based method incurs significantly higher computational costs than the others, which limits its practicality on large-scale datasets.

Hyperparameters analysis

In this section, we evaluate the impact of hyperparameters on model detection performance. In the proposed contrastive learning loss function, the hyperparameter β adjusts the degree of clustering between anchor samples and positive samples when constructing negative sample pairs, while the temperature coefficient τ the distinction between positive and negative samples. A smaller β results in tighter clustering between the representation vectors of positive samples generated by the encoder but reduces the distinction from negative sample representations. Conversely, a smaller τ increases the distinction between the representation vectors of positive and negative samples but decreases the clustering among positive sample representations. Therefore, we assess the effect of different combinations of β and τ on model performance. The results are shown in Table 12 and Table 13

From Tables Table 12 and Table 13, it can be seen that the hyperparameter β has a greater impact on model performance than τ. When β is small, the distinction between the representation vectors of different classes is reduced, resulting in a drop in accuracy. Specifically, on the UNSW-NB15 dataset, the number of malicious samples is nearly twice that of normal samples. Due to the design of the BGMM-based detection algorithm, samples are more likely to be classified as malicious. For example, when $\beta = 0.05$ and $\tau = 0.15$, all samples are classified as malicious, leading to a recall close to 100%. In contrast, on the Distrinet-CIC-IDS2017 dataset, the number of normal samples far exceeds the number of malicious samples, making it more likely for samples to be classified as normal. For instance, when $\beta = 0.05$ and $\tau = 0.15$, some malicious samples are misclassified as normal, resulting in lower recall.

Furthermore, selecting appropriate values for β and τ helps maintain tight clustering among the representation

Table 10 Recall rate(%) of each type between ours and baselines on the Distrinet-CIC-IDS2017 dataset

Type	Method									
	Ours	**AOC-IDS**	**CIDS**	**CANET**	**CTS**	**SVM**	**DTC**	**RF**	**XGBoost**	**AdaBoost**
Normal	99.70	95.33	99.37	**99.95**	99.44	99.76	99.70	99.89	99.72	99.76
Botnet	85.90	**100.00**	96.79	3.85	1.28	5.77	0.00	1.28	0.00	89.74
DDoS	99.99	**100.00**	99.95	99.36	99.96	99.94	99.16	99.97	99.99	99.82
DoS GoldenEye	98.21	**99.36**	97.85	93.56	0.86	77.52	92.13	90.05	95.35	96.13
DoS Hulk	99.91	**100.00**	99.52	98.58	99.82	99.71	99.63	99.90	99.95	99.69
DoS Slowhttptest	97.12	**99.71**	93.66	84.73	88.76	54.76	16.14	58.21	77.52	93.66
DoS Slowloris	85.51	**100.00**	84.12	77.65	93.90	79.16	45.24	79.67	93.52	99.75
FTP-Patator	**100.00**	**100.00**	99.74	84.69	99.74	0.00	99.49	99.62	99.62	99.87
Heartbleed	25.00	**100.00**	0.00	0.00	0.00	25.00	50.00	25.00	**100.00**	0.00
Infiltration	0.00	**100.00**	0.00	0.00	25.00	25.00	50.00	0.00	0.00	0.00
Infiltration -Portscan	93.25	**99.64**	57.43	5.31	92.89	0.00	82.90	80.56	81.01	83.62
Portscan	98.20	**99.40**	80.48	4.80	90.09	4.50	4.50	10.51	34.53	61.56
SSH-Patator	99.19	**100.00**	99.19	98.54	98.38	0.16	0.00	0.00	98.06	98.87
Brute Force	**92.31**	**92.31**	**92.31**	**92.31**	**92.31**	0.00	0.00	0.00	7.69	0.00
Injection	0.00	**100.00**	66.67	0.00	0.00	0.00	0.00	0.00	0.00	33.33
XSS	**100.00**	**100.00**	**100.00**	0.00	**100.0**	0.00	0.00	0.00	**100.00**	0.00

The optimal metric performance and the minority class are highlighted in bold

Table 11 Computational cost(\s) comparison between ours and baselines on the Distrinet-CIC-IDS-2017 dataset

Method	Computational cost	
	T_{train}	T_{test}
Ours	531.89	0.60
AOC-IDS	9372.37	6.65
CIDS	1152.48	2.07
CANET	880.49	21.35
CTS	1520.48	35.72
SVM	25620.45	2979.58
DTC	31.64	**0.04**
RF	431.76	1.64
AdaBoost	234.79	0.68
XGBoost	**29.39**	0.18

The optimal metric performance is highlighted in bold

vectors of normal samples while preserving separation from malicious sample vectors, allowing the model to achieve optimal performance.

For the BGMM-based detection algorithm, we evaluate how varying the number of initial components affects model detection performance. A smaller initial number of components in BGMM helps reduce model complexity and training time. The results are presented in Table 14 and Table 15. As shown in the tables, on the UNSW-NB15 dataset, where there are fewer sample categories, model performance stabilizes with an initial component count of 10. In contrast, on the Distrinet-CIC-IDS2017 dataset, which has a broader variety of sample categories, a higher initial component count, such as 20, improves model performance.

Ablation experiments

In this section, ablation experiments are conducted to verify the effectiveness of each proposed module. Specifically, the proposed loss function based on contrastive learning is first replaced with a binary cross-entropy loss function, denoted as "w/o contrastive loss". When trained with the cross-entropy loss function, the model directly outputs the predict results. Then, to validate the effectiveness of the detection method, the detection method proposed by Zhang et al. (2024), which employs two Gaussian distributions for the detection, is experimented with, denoted as "w/o BGMM". Finally, neither Beyesian Gaussian Mixture Model nor Gaussian distribution is used, instead, a fixed threshold is employed for the detection, denoted as "w/ fixed threshold". In this paper, the threshold is set to the top 5% of the cosine similarity scores between all normal representation vectors and the normal template. The results are as shown in Table 16, Table 17.

The results indicate that the proposed loss function, which realizes higher accuracy and F1-score both on the UNSW-NB15 dataset and Distrinet-CIC-IDS2017 dataset, outperforms the binary cross-entropy loss function. Comparing to the binary cross-entropy loss function, the loss function based on contrastive learning enables the network focus on learning the differences between

Table 12 Performance (%) of the proposed method with different hyperparameter combinations on the UNSW-NB15 dataset

	τ	Acc	Pre	Rec	F1
$\beta = 0.05$	0.05	88.28	84.22	96.88	90.10
	0.075	88.48	85.50	95.22	90.10
	0.1	55.06	55.06	**100.00**	71.02
	0.125	55.06	55.06	**100.00**	71.02
	0.15	55.06	55.06	**100.00**	71.02
$\beta = 0.075$	0.05	81.91	75.32	99.87	85.88
	0.075	86.17	80.76	98.30	88.67
	0.1	88.40	84.03	97.46	90.25
	0.125	88.67	85.08	96.31	90.35
	0.15	**91.26**	**89.74**	95.01	**92.30**
$\beta = 0.1$	0.05	80.97	74.32	99.98	85.26
	0.075	81.46	74.86	99.87	85.58
	0.1	83.43	77.07	99.50	86.86
	0.125	85.32	79.94	97.91	88.01
	0.15	86.61	81.36	98.17	88.98
$\beta = 0.125$	0.05	81.05	74.40	99.99	85.32
	0.075	81.08	74.43	99.99	85.34
	0.1	81.16	74.52	99.91	85.39
	0.125	81.49	74.90	99.84	85.59
	0.15	81.94	75.36	99.84	85.89
$\beta = 0.15$	0.05	81.01	74.36	99.99	85.29
	0.075	81.05	74.40	99.99	85.31
	0.1	80.97	74.32	99.98	85.26
	0.125	81.10	74.45	99.98	85.35
	0.15	81.04	74.40	99.96	85.30

The optimal metric performance is highlighted in bold

Table 13 Performance (%) of the proposed method with different hyperparameter combinations on the Distrinet-CIC-IDS2017 dataset

	τ	Acc	Pre	Rec	F1
$\beta = 0.05$	0.05	97.81	92.19	97.00	94.53
	0.075	96.95	89.37	95.77	92.46
	0.1	97.36	92.97	93.53	93.25
	0.125	97.23	95.52	90.04	92.70
	0.15	97.59	96.98	90.50	93.63
$\beta = 0.075$	0.05	98.83	94.69	**99.63**	97.09
	0.075	98.88	95.27	99.22	97.20
	0.1	98.57	94.67	98.21	96.41
	0.125	98.69	98.03	95.18	96.58
	0.15	98.48	97.32	94.84	96.07
$\beta = 0.1$	0.05	99.61	98.87	99.16	99.01
	0.075	99.60	98.82	99.15	98.98
	0.1	99.47	98.95	98.33	98.64
	0.125	99.52	**99.14**	98.37	98.75
	0.15	99.50	99.00	98.42	98.71
$\beta = 0.125$	0.05	99.60	99.03	98.91	98.97
	0.075	99.46	97.66	99.61	98.62
	0.1	99.58	98.43	99.41	98.92
	0.125	99.55	98.34	99.39	98.86
	0.15	99.35	98.00	98.71	98.35
$\beta = 0.15$	0.05	99.54	98.93	98.71	98.82
	0.075	**99.66**	98.77	99.47	**99.11**
	0.1	99.60	98.85	99.10	98.98
	0.125	99.63	98.63	99.50	99.06
	0.15	99.55	98.31	99.42	98.86

The optimal metric performance is highlighted in bold

normal class and malicious classes, as well as the similarities within normal class, which in aid of constructing a robust normal pattern for subsequent detection, rather than predicting the labels of samples. The adjustment is more in line with the actual situation of the intrusion detection task, rather than just treating it as a classification task, and thus can improve the overall performance of the model.

The proposed detection algorithm based on BGMM demonstrates more robust performance compared to other methods. Unlike the "without BGMM" approach, BGMM can automatically determine the appropriate number of effective components based on the data distribution, eliminating the need for prior knowledge to specify the number of components. This capability allows BGMM to fit the data more accurately, resulting in higher accuracy and F1-score, which confirms its effectiveness and practicality in addressing complex network attacks. In contrast, the fixed threshold detection method, while simpler, lacks flexibility. Moreover, finding

Table 14 Performance (%) with initial BGMM components on the UNSW-NB15 dataset

component	Acc	Pre	Rec	F1
10	**91.27**	**89.74**	**95.01**	**92.30**
20	91.26	89.73	**95.01**	92.29
30	91.26	89.73	**95.01**	92.29

The optimal metric performance is highlighted in bold

Table 15 Performance (%) with initial BGMM components on the Distrinet-CIC-IDS2017 dataset

component	Acc	Pre	Rec	F1
10	98.40	92.47	**99.93**	96.06
20	**99.66**	**98.77**	99.47	**99.11**
30	99.64	98.63	99.55	99.09

The optimal metric performance is highlighted in bold

Table 16 Ablation experiments on the UNSW-NB15 dataset

Method	Acc	Pre	Rec	F1
Ours	**91.27**	**89.74**	95.01	**92.30**
w/o contrastive loss	83.83	77.80	98.82	87.06
w/o BGMM	81.96	75.50	**99.54**	85.87
w/ fixed threshold	89.79	86.45	96.60	91.24

The optimal metric performance is highlighted in bold

Table 17 Ablation experiments on the Distrinet-CIC-IDS2017 dataset

Method	Acc	Pre	Rec	F1
Ours	**99.66**	**98.77**	99.47	**99.12**
w/o contrastive loss	97.34	92.50	93.99	93.24
w/o BGMM	87.47	60.92	**99.99**	75.71
w/ fixed threhold	95.96	82.87	99.98	90.62

The optimal metric performance is highlighted in bold

an optimal threshold that maximizes model performance is challenging.

Interpretability exploration

In this section, we conduct a brief exploration of the proposed method's interpretability. To validate the effectiveness of the contrastive learning loss function, we randomly sample 5,000 data points from the dataset and use the trained encoder to obtain their feature representations. These representations are then reduced to two dimensions using t-SNE for visualization, as shown in the Fig. 3a and b. The visualization shows that, compared to the original samples, the representation vectors of normal samples form tighter clusters and display a noticeable separation from those of malicious samples. It is noteworthy that in the UNSW-NB15 dataset, certain attack patterns exhibit similarities to normal traffic, which may lead to overlaps in the representation vectors across different classes.

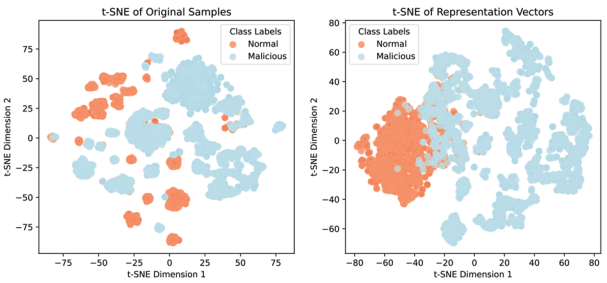
Furthermore, we utilize histograms to visualize the cosine similarity scores between the representation vectors and the normal template, thereby elucidating the BGMM-based detection algorithm employed, as illustrated in the Figs. 4 and 5. The figures show that the cosine similarity scores distribution of the normal pattern is concentrated, while the cosine similarity distribution scores of the malicious pattern is more dispersed. The use of multiple components in the BGMM allows for a more precise fitting of the distributions corresponding to different patterns, which helps improve the model's robustness when facing complex traffic.

Dissusion and limitations

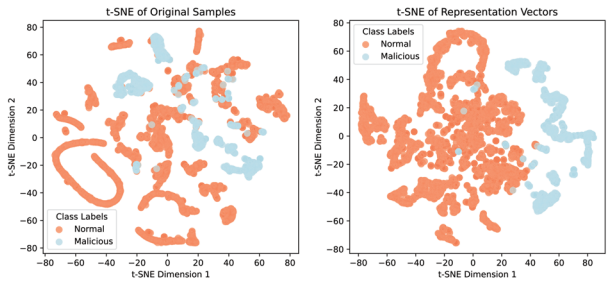
In this section, the current main research directions and challenges in the field of network intrusion detection are briefly introduced. According to the detection process, methods can be categorized into one-stage detection methods and two-stage detection methods.

One-stage detection methods treat the intrusion detection task as a classification task, with the model architecture typically combining a feature extraction network and a classification head. The advantage of these methods is that they directly provide classification results without additional processing. However, they are limited by the class imbalance problem, i.e., the trained models tend to favor the class corresponding to the majority of samples(usually normal class), which can result in lower sensitivity, particularly for detecting rare or sophisticated attacks, as the model is more likely to classify them as normal traffic.

The processing flow of two-stage intrusion detection methods usually involves using a feature extraction network to construct the normal pattern first, followed by designing an algorithm for the detection. The advantage



(a) t-SNE visualization on the UNSW-NB15 dataset



(b) t-SNE visualization on the Distrinet-CIC-IDS2017 dataset

Fig. 3 t-SNE visualizations on two datasets

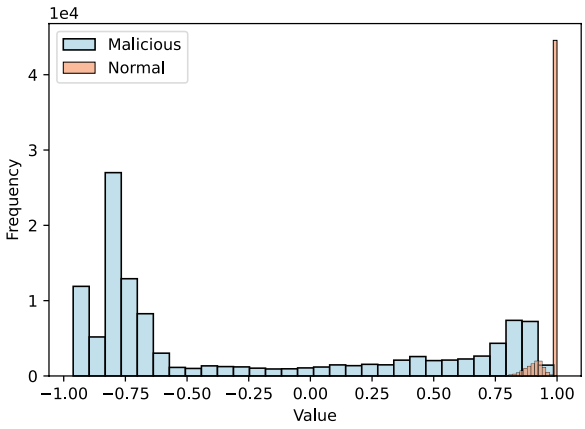


Fig. 4 Histogram on the UNSW-NB15 dataset

of these two-stage methods lie in their ability to improve both sensitivity and specificity. By focusing on building a clear model of normal behavior in the first stage, the system becomes more effective at identifying deviations that indicate potential attacks in the second stage. This separation allows each stage to specialize and optimize for its specific task, reducing the risk of false negatives and improving overall detection performance, especially for rare or sophisticated attacks. However, the disadvantage is that each stage requires careful design, resulting in a heavier workload compared to one-stage methods.

Conclusion

In this paper, an innovative network intrusion detection method consisting of a representation module based on contrastive learning and a detection module based on Bayesian Gaussian Mixture Model is proposed. Specifically, a novel contrastive learning loss function is proposed to enhance the representation capability of the network, and Bayesian Gaussian Mixture Model is utilized for anomaly detection to cope with complex and varied network attacks. The proposed method neither requires complex feature engineering nor oversampling for minority classes, which makes it more practical. The rationality of the proposed method is explained through detailed theoretical analysis, and the performance is evaluated by various experiments on two benchmark datasets. The results show that, compared to the SOTA method, the proposed method achieves 91.27% accuracy and 92.30% F1-score on the UNSW-NB15 dataset, reflecting improvements of 1.85% and 2.35%. On the Distrinet-CIC-IDS2017 dataset, it achieves 99.66% accuracy and 99.12% F1-score, representing improvements of 0.05% and 0.12%. These results highlight the effectiveness and advancement of the proposed method.

As mentioned above, contrastive learning excels at capturing the differences between different classes and the similarities within the same class, making it well-suited for the construction of the normal pattern. While our approach has demonstrated promising results, there is still significant potential in exploring other contrastive learning methods to enhance anomaly detection. Specifically, techniques such as multi-task contrastive loss(Viana et al. 2023), self-supervised contrastive learning(Jaiswal et al. 2020), and instance-level contrastive learning(Liu et al. 2023a) could offer more refined feature representations. These methods are better equipped to capture subtle differences between normal and abnormal traffic, improving the model's ability to adapt and perform well across a broader range of attack scenarios. Therefore, in future work, we plan to investigate these advanced contrastive learning approaches to further improve the model's detection capabilities.

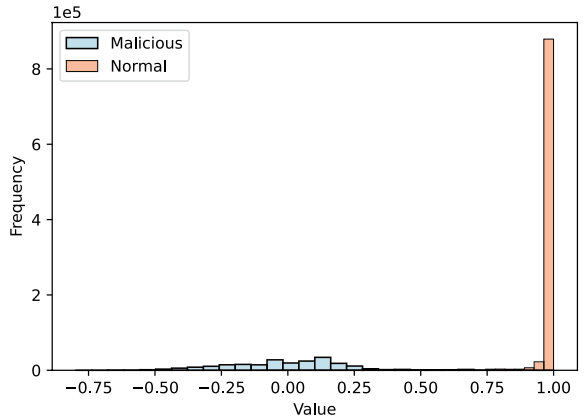


Fig. 5 Histogram on the distrinet-CIC-IDS2017 dataset

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Data will be made available on request.

Declarations

Competing interests

The authors declare that they have no competing interests.

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References

Abdelkhalek A, Mashaly M (2023) Addressing the class imbalance problem in network intrusion detection systems using data resampling and deep learning. J Supercomput 79(10):10611–10644. https://doi.org/10.1007/s11227-023-05073-x

Ahmad Z, Shahid Khan A, Wai Shiang C et al (2021) Network intrusion detection system: a systematic study of machine learning and deep learning approaches. Trans Emerg Telecommun Technol 32(1):e4150. https://doi.org/10.1007/s11227-023-05073-x

Albulayhi K, Abu Al-Haija Q, Alsuhibany SA et al (2022) Iot intrusion detection using machine learning with a novel high performing feature selection method. Appl Sci 12(10):5015. https://doi.org/10.3390/app12105015

Altunay HC, Albayrak Z (2023) A hybrid cnn+ lstm-based intrusion detection system for industrial iot networks. Eng Sci Technol Int J 38:101322. https://doi.org/10.1016/j.jestch.2022.101322

An C, Feng J, Lv K et al (2022) Cont: contrastive neural text generation. Adv Neural Inform Process Syst 35:2197–2210 https://doi.org/10.48550/arXiv.2205.14690

Andresini G, Appice A, De Rose L et al (2021) Gan augmentation to deal with imbalance in imaging-based intrusion detection. Futur Gener Comput Syst 123:108–127. https://doi.org/10.1016/j.future.2021.04.017

Balyan AK, Ahuja S, Lilhore UK et al (2022) A hybrid intrusion detection model using ega-pso and improved random forest method. Sensors 22(16):5986. https://doi.org/10.3390/s22165986

Bishop CM, Nasrabadi NM (2006) Pattern recognition and machine learning. Springer, Cham

Blei DM, Kucukelbir A, McAuliffe JD (2017) Variational inference: a review for statisticians. J Am Stat Assoc 112(518):859–877. https://doi.org/10.1080/01621459.2017.1285773

Dai W, Li X, Ji W et al (2024) Network intrusion detection method based on cnn, bilstm, and attention mechanism. IEEE Access. https://doi.org/10.1109/ACCESS.2024.3384528

Dempster AP, Laird NM, Rubin DB (1977) Maximum likelihood from incomplete data via the em algorithm. J Roy Stat Soc: Ser B (Methodol) 39(1):1–22. https://doi.org/10.1111/j.2517-6161.1977.tb01600.x

Disha RA, Waheed S (2022) Performance analysis of machine learning models for intrusion detection system using gini impurity-based weighted random forest (giwrf) feature selection technique. Cybersecurity 5(1):1. https://doi.org/10.1186/s42400-021-00103-8

Fu Y, Du Y, Cao Z et al (2022) A deep learning model for network intrusion detection with imbalanced data. Electronics 11(6):898. https://doi.org/10.3390/electronics11060898

Gao T, Yao X, Chen D (2021) Simcse: Simple contrastive learning of sentence embeddings. arXiv preprint arXiv:2104.08821https://doi.org/10.18653/v1/2021.emnlp-main.552

Guezzaz A, Benkirane S, Azrour M et al (2021) A reliable network intrusion detection approach using decision tree with enhanced data quality. Security Commun Netw 1:1230593. https://doi.org/10.1155/2021/1230593

Halbouni A, Gunawan TS, Habaebi MH et al (2022) Cnn-lstm: hybrid deep neural network for network intrusion detection system. IEEE Access 10:99837–99849. https://doi.org/10.1109/ACCESS.2022.3206425

Halim Z, Yousaf MN, Waqas M et al (2021) An effective genetic algorithm-based feature selection method for intrusion detection systems. Comput Security 110:102448. https://doi.org/10.1016/j.cose.2021.102448

He J, Wang X, Song Y et al (2023) A multiscale intrusion detection system based on pyramid depthwise separable convolution neural network. Neurocomputing 530:48–59. https://doi.org/10.1016/j.neucom.2023.01.072

Hu J, Liu C, Cui Y (2021) An improved cnn approach for network intrusion detection system. Int J Netw Security 23(4):569–575

Huang S, Lei K (2020) Igan-ids: an imbalanced generative adversarial network towards intrusion detection system in ad-hoc networks. Ad Hoc Netw 105:102177. https://doi.org/10.1016/j.adhoc.2020.102177

Jaiswal A, Babu AR, Zadeh MZ et al (2020) A survey on contrastive self-supervised learning. Technologies 9(1):2. https://doi.org/10.3390/technologies9010002

Jiang H, He Z, Ye G et al (2020) Network intrusion detection based on pso-xgboost model. IEEE Access 8:58392–58401. https://doi.org/10.1109/ACCESS.2020.2982418

Jin S, Chung JG, Xu Y (2021) Signature-based intrusion detection system (ids) for in-vehicle can bus network. In: 2021 IEEE international symposium on circuits and systems (ISCAS), IEEE, pp 1–5, https://doi.org/10.1109/ISCAS51556.2021.9401087

Kasongo SM, Sun Y (2020) Performance analysis of intrusion detection systems using a feature selection method on the unsw-nb15 dataset. J Big Data 7(1):105. https://doi.org/10.1186/s40537-020-00379-6

Kilichev D, Turimov D, Kim W (2024) Next-generation intrusion detection for iot evcs: integrating cnn, lstm, and gru models. Mathematics 12(4):571. https://doi.org/10.3390/math12040571

Lee J, Park K (2021) Gan-based imbalanced data intrusion detection system. Pers Ubiquit Comput 25(1):121–128. https://doi.org/10.1007/s00779-019-01332-y

Li J, Othman MS, Chen H et al (2024) Optimizing iot intrusion detection system: feature selection versus feature extraction in machine learning. J Big Data 11(1):36

Li L, Lu Y, Yang G et al (2024) End-to-end network intrusion detection based on contrastive learning. Sensors 24(7):2122. https://doi.org/10.3390/s24072122

Li S, Xia X, Ge S, et al (2022) Selective-supervised contrastive learning with noisy labels. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 316–325, https://doi.org/10.48550/arXiv.2203.04181

Liu C, Antypenko R, Sushko I et al (2022) Intrusion detection system after data augmentation schemes based on the vae and cvae. IEEE Trans Reliab 71(2):1000–1010. https://doi.org/10.1109/TR.2022.3164877

Liu C, Wen J, Luo X, et al (2023) Dicnet: Deep instance-level contrastive network for double incomplete multi-view multi-label classification. In: Proceedings of the AAAI conference on artificial intelligence, pp 8807–8815, https://doi.org/10.1609/aaai.v37i7.26059

Liu L, Engelen G, Lynar T, et al (2022b) Error prevalence in nids datasets: A case study on cic-ids-2017 and cse-cic-ids-2018. In: 2022 IEEE Conference on Communications and Network Security (CNS), IEEE, pp 254–262, https://doi.org/10.1109/CNS56114.2022.9947235

Mahbooba B, Timilsina M, Sahal R et al (2021) Explainable artificial intelligence (xai) to enhance trust management in intrusion detection systems using decision tree model. Complexity 1:6634811. https://doi.org/10.1155/2021/6634811

Moustafa N, Slay J (2015) Unsw-nb15: a comprehensive data set for network intrusion detection systems (unsw-nb15 network data set). In: 2015 military communications and information systems conference (MilCIS), IEEE, pp 1–6, https://doi.org/10.1109/MilCIS.2015.7348942

Ren K, Yuan S, Zhang C et al (2023) Canet: a hierarchical cnn-attention model for network intrusion detection. Comput Commun 205:170–181. https://doi.org/10.1016/j.comcom.2023.04.018

Sharafaldin I, Lashkari AH, Ghorbani AA et al (2018) Toward generating a new intrusion detection dataset and intrusion traffic characterization. ICISSp 1:108–116

Sommestad T, Holm H, Steinvall D (2022) Variables influencing the effectiveness of signature-based network intrusion detection systems. Inform Security J Global Perspective 31(6):711–728. https://doi.org/10.1080/19393555.2021.1975853

Su T, Sun H, Zhu J et al (2020) Bat: deep learning methods on network intrusion detection using nsl-kdd dataset. IEEE Access 8:29575–29585. https://doi.org/10.1109/ACCESS.2020.2972627

Tang B, Lu Y, Li Q et al (2023) A diffusion model based on network intrusion detection method for industrial cyber-physical systems. Sensors 23(3):1141. https://doi.org/10.3390/s23031141

Turukmane AV, Devendiran R (2024) M-multisvm: an efficient feature selection assisted network intrusion detection system using machine learning. Comput Security 137:103587. https://doi.org/10.1016/j.cose.2023.103587
Viana TB, Souza VL, Oliveira AL et al (2023) A multi-task approach for contrastive learning of handwritten signature feature representations. Expert Syst Appl 217:119589. https://doi.org/10.1016/j.eswa.2023.119589
Wang N, Chen Y, Hu Y, et al (2022a) Feco: Boosting intrusion detection capability in iot networks via contrastive learning. In: IEEE INFOCOM 2022-IEEE Conference on Computer Communications, IEEE, pp 1409–1418, https://doi.org/10.1109/INFOCOM48880.2022.9796926
Wang Y, Li J, Wang H, et al (2022b) Wav2vec-switch: Contrastive learning from original-noisy speech pairs for robust speech recognition. In: ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, pp 7097–7101, https://doi.org/10.48550/arXiv.2110.04934
Wu C, Li W (2021) Enhancing intrusion detection with feature selection and neural network. Int J Intell Syst 36(7):3087–3105. https://doi.org/10.1002/int.22397
Yang Z, Liu X, Li T et al (2022) A systematic literature review of methods and datasets for anomaly-based network intrusion detection. Comput Security 116:102675. https://doi.org/10.1016/j.cose.2022.102675
Yin Y, Jang-Jaccard J, Xu W et al (2023) Igrf-rfe: a hybrid feature selection method for mlp-based network intrusion detection on unsw-nb15 dataset. J Big Data 10(1):15. https://doi.org/10.1186/s40537-023-00694-8
Yuan L, Sun J, Zhuang S, et al (2024) Manticore: An unsupervised intrusion detection system based on contrastive learning in 5g networks. In: ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, pp 4705–4709, https://doi.org/10.1109/ICASSP48485.2024.10447814
Yue Y, Chen X, Han Z et al (2022) Contrastive learning enhanced intrusion detection. IEEE Trans Netw Serv Manage 19(4):4232–4247. https://doi.org/10.1109/TNSM.2022.3218843
Zhang H, Huang L, Wu CQ et al (2020) An effective convolutional neural network based on smote and gaussian mixture model for intrusion detection in imbalanced dataset. Comput Netw 177:107315. https://doi.org/10.1016/j.comnet.2020.107315
Zhang X, Zhao R, Jiang Z, et al (2024) Aoc-ids: Autonomous online framework with contrastive learning for intrusion detection. arXiv preprint arXiv:2402.01807. https://doi.org/10.48550/arXiv.2402.01807
Zhou Y, Cheng G, Jiang S et al (2020) Building an efficient intrusion detection system based on feature selection and ensemble classifier. Comput Netw 174:107247. https://doi.org/10.1016/j.comnet.2020.107247
Zimek A, Schubert E, Kriegel HP (2012) A survey on unsupervised outlier detection in high-dimensional numerical data. Stat Anal Data Mining ASA Data Sci J 5(5):363–387. https://doi.org/10.1002/sam.11161

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